

Charotar University of Science and Technology (CHARUSAT)
Faculty of Technology and Engineering (FTE)

Subject: Advanced Web Development Frameworks (ITUE301)

Semester: 5th

Teaching Scheme:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/Week	-	2	-	2	2
Marks	-	50	-	50	

Course Pre-requisites:

- Database concepts, Query optimization, Schema Design, Indexing
- Scripting language Features.

Course Outcomes:

By the end of the course, students will be able to:

C01: Design and Implement component-based frontend systems using reactive state management and performance optimization strategies for scalable web applications.

C02: Design and Implement secure RESTful backend services incorporating middleware pipelines, validation layers, and caching mechanisms.

C03: Select and evaluate appropriate data modeling, migration, encryption, and storage strategies based on workload characteristics and consistency requirements.

C04: Design scalable web system architectures by applying stateless service design, load balancing, asynchronous processing, and performance profiling techniques.

C05: Implement cloud-native deployment pipelines using containerization, CI/CD automation, and deployment strategies while monitoring reliability and resource optimization.

C06: Apply security engineering principles and responsibly integrate AI-based services into web systems while analyzing ethical, privacy, and governance implications.

➤ **Pre-requisites online video / course:**

- Client-Server Architecture
- UI/UX Principles
- State Management
- Schema design principles
- ACID Vs BASE properties

Syllabus:

Unit No.	Topic	Hrs.
1	Frontend and Component Architecture	5
	• Overview of Frontend architecture: MPA, SPA, SSR, SSG • Design Methodology: Responsive, Atomic • Overview: UI Component Libraries • Introduction: Component Architecture • State Management: Reactive Patterns • Routing Concepts • API Integration and Consumption strategies	
2	Frontend Engineering	4
	• Rendering Lifecycle • Lazy Code Loading • WCAG Principles • Overview: PWA • Performance Optimization • Frontend Security (XSS, CSRF, Token Handling) and Automated Testing (Application level and Security level)	
3	Backend Architecture and Data Engineering	9
	• Maturity Model: Richardson • Middleware and Request Lifecycle • RESTful API Design • Data Validation Layers and Error Handling • Schema Enforcement • In-memory Caching Strategies • Secure Storage Principles • Database Migration Principles • Overview: Data Encryption	
4	Scalable Web Systems	11
	• Architecture: Monolithic Vs Microservices • Connection Pooling and Query Caching • Scaling: Horizontal Vs Vertical • Stateless Services	

	• Load Balancing and Fault Tolerance • Response Time and Resource Utilization: Minimization • CDN Fundamentals • Asynchronous Processing and Message Queues - Background Data Processing, Notification • Profile: CPU and Memory Usage, Energy-efficient web systems, Data Transfer Optimization • Performance Load Testing – RPS, Latency, Error Rate • Site Reliability Engineering	
5	Cloud Deployment • Overview: Cloud Computing, Cloud Native Systems • Introduction: Cloud Service Models – Serverless, IaaS, PaaS • Containerization and OCI Standards • Docker Fundamentals and Docker File Optimization • CI/CD Pipeline: Automated Test Execution, Performance Degradation • Overview: Resource Limits Configuration • Deployment Strategies: Blue-Green, Rolling Updates	7
6	Security and AI Integration • Introduction: Threat Model • Advanced Authentication and Authorization • Overview: HTTPS and TLS • Secure Coding: Introduction and Practices • AI Service Integration and Prompt Engineering (Scalability) • AI-assisted Customized Designing • Introduction: Responsible AI	4

Recommended Study Material:

❖ Text books:

1. *Web Technologies* by Uttam K. Roy, Oxford University Press
2. *Internet and Web Technologies* by Raj Kamal, McGraw Hill Education
3. *Cloud Computing* by Anthony T. V., et. al, McGraw Hill Education

❖ Reference Books:

1. *Designing Data-Intensive Applications* by Martin Kleppmann
2. *Clean Code* by Robert C. Martin
3. *Distributed Systems: Concepts and Design* by George C., et.al., Pearson Education

Instructions to Subject Teachers:

1. Selection of tools/databases/scripting languages/platform is under the purview of the subject teachers. However, Installation and configuration of tools to be used must be demonstrated.
2. Teaching Pedagogy, Continuous evaluation as per NEP-2020 guidelines, and Practical List

must be prepared in advanced according to Open Book Examination as well as the Course Outcomes addressing specific skills, and must be included in the Lesson Plan, which should be made available to students before commencement of the semester.

3. Practical Problem Definition must have pre and post laboratory exercise/questions addressing to specific higher-order skills must be clearly mentioned apart from resources to be utilized for the solution. Remembering nature questions are strictly prohibited.
4. Students must demonstrate performance benchmarking for related topics.
5. Appropriate tool may be demonstrated to prepare web related project documentation.
6. At least two expert sessions each of four hours' duration must be conducted addressing advanced features of selected and/or industry demand tools. Prior approval process must be carried out.
7. Selection of Reference books and Online resources like COURSERA, NPTEL, etc. is under purview of the subject teachers.
8. Subject teachers should announce at least one COURSERA/NPTEL certification course inline with syllabus as an advanced knowledge (not exactly similar to syllabus).